SPACE X · WHOLE COURSE · v1

All seven sections on one page

One sentence to say, one number to know, and one thing the instructor said that you should not repeat. If you can say the middle column from memory, you can hold the conversation.

Section	Say this	Number to know	Don't repeat
1 · Mission design	An orbit is falling and missing. Low is fast and sharp; one exact height hangs still over the equator.	LEO 400 km: 92 min, 7.7 km/s · GEO 35,786 km	"36,000 km from the centre" (that is the altitude)
2 · Space environment	The air did five jobs for you: sunscreen, pressure, damper, fan, ground wire. Space does none.	atomic oxygen rules 200–600 km	"atomic oxygen appears at 320 km"
3 · Orbital mechanics	Six numbers describe any orbit. The lazy half-ellipse is cheapest; a flyby steals speed from a planet.	Mars: 259 days, window every 26 months	"Lambert takes the Δv and gives the time" (reversed)
4 · Propulsion	Every engine throws mass backward. Chemical pushes hard and is thirsty; electric is gentle and efficient; nuclear never flew.	$F = \dot{m} \times v_e$ · Isp 450 s / 3,000 s / 900 s	"nuclear is efficient because it is hotter" (lighter exhaust)
5 · Communications	Write the message on a radio wave, aim it with a dish, send it through the radio window, land louder than the noise.	$c = \lambda \times f$ · Cassini: 20 W, 10^{-16} W received	"a dish adds power" · "k is about wavelength"
6 · Attitude control	Sense, compare, twist, forever. Gyros drift and stars fix them. Wheels swap spin; jets and magnets remove it.	star tracker 0.001° · Hubble 0.007 arcsec	"Explorer 1 failed" · "inertia = mass in a direction"
7 · Power	A sunlight sandwich near the Sun, a bucket for the dark, a warm rock where the Sun doesn't reach.	1,361 W/m² at Earth · cells 30% · Pu-238 87.7 years	"three layers = 3× efficiency" · "RTGs last 500 years"

The thread: a spacecraft has to do everything for itself. Get there (1, 3, 4) · survive (2) · talk (5) · point (6) · stay powered (7).

The four formulas worth carrying

$F = \dot{m} \times v_e$ · thrust = mass thrown per second × exhaust speed

$c = \lambda \times f$ · wavelength × frequency = speed of light

P = flux × area × efficiency · solar array power

battery = load × shadow time ÷ depth of discharge

The one idea that repeats three times

Twice as far = a quarter as much. It is why gravity weakens with height (Section 1), why a radio signal fades (Section 5) and why sunlight fades (Section 7). Anything that spreads out over a sphere does this.

SPACE X · PLAN FOR TOMORROW

About three hours, in five blocks

Sections 1–3 you have rehearsed before. Sections 4–7 you have only copied. So the time goes to 4–7, and most of it to saying things out loud.

The blocks

Time	Do this
40 min	**Read.** For each of the 12 lectures in Sections 4–7: the "Like you're 5" box and the "Where the instructor was loose" list. Nothing else. About 3 minutes each.
40 min	**Quiz.** App → Quiz → new questions. Scenes 8 to 19. Every miss, read the explanation out loud.
45 min	**Talk.** App → Teach. Section 4, 5, 6 and 7 talks, once each, timer running, cue card hidden after the first minute. Send me any one of them.
20 min	**Whole course.** App → Teach → Whole course, twice. The second time, no cue card.
20 min	**Quiz 5** in Udemy (designing a solar array), using the recipe on the right.

If you only have one hour: the page before this one, then the whole-course talk twice, then the Section 4–7 "loose" lists.
If it is a test, not a talk: swap the talk block for a second quiz block and the Numbers tab.

Recipe for Quiz 5: size a solar array

1. Sunlight per m² = 1,361 ÷ (distance in AU)².
2. Usable per m² = that × efficiency × any loss factor they give.
3. Area = power needed ÷ usable per m².
4. If there is a battery to refill, add its recharge power to "power needed" first.
Example: 500 W at Mars, 30% cells, no other losses. 1,361 ÷ 1.52^2 = 590. 590 × 0.30 = 177 W per m². 500 ÷ 177 = **2.8 m²**.

What is still open

You have not explained anything back to me since Section 3. That is 16 lectures of copying with no check. One talk sent tomorrow is worth more than re-reading everything.

Paused during the push, yours to ask for afterwards: concept-card pages for Sections 6–7, the copy-into-notes pages, and the Who's Who names for Sections 6–7.